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Tennis serve volume, distribution and accelerometer load during training and tournaments from wearable microtechnology

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ABSTRACT

This study analysed the volume, intensity and distribution of tennis serving between (1) training and tournament mesocycles, and (2) within-mesocycle based on session-type, drill-type and week. Seven elite females wore a GPS device, with in-built accelerometer and gyroscope, during training and tournament mesocycles. Sessions were classified as “practice sessions” or “official matches” and drills as “serve-focused drills” or “point-play”. Training weeks were classified according to their timing prior to tournaments (“Week –3”, “Week –2” and “Week –1”). Machine learning algorithms detected serves from the raw accelerometer and gyroscope data to report serve count, load and rotation magnitude in absolute and relative (min^{-1} or serve) values. Cohen’s d with 95% confidence intervals compared serve volume (count), intensity (serve load, rotation magnitude) and distribution (n.min^{-1}) between session- and drill-type during mesocycles. Cumulative serve counts were compared between training weeks. Increased absolute serve load was observed during official matches compared to practice sessions ($d = 0.97\text{--}1.07[0.69\text{--}1.30]$). Serve-focused drills in training mesocycles have lower absolute counts and serve load ($d = 0.82[0.54\text{--}1.34]$), though greater relative count and load ($d = 1.10\text{--}1.87[0.59\text{--}2.19]$). Trivial-small ES existed between training weeks for cumulative serve count ($d = 0.12\text{--}0.25[-0.83\text{--}1.27]$). Session- and drill-level analyses highlight greater mechanical demands in official matches, though lower serve density in point-play.

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Microsensors; training prescription; accelerometry; racquet sports

1. Introduction

The serve is a key determinant of match-play success in tennis (Brown, 2021; O’Donoghue & Ingram, 2001) and features heavily in the training programmes of high-performance players (Fett et al., 2017; Fitzpatrick et al., 2019; Krause et al., 2019). However, empirically supported recommendations on the volume, intensity and distribution of serving practice for elite players are limited. In part this is due to the dispersed training exposures available for players around regular tournament play (Kovacs, 2019; Reid et al., 2009). As such, appropriate serve training loads (i.e. volume and intensity)

within both dedicated training and competitive periods may ensure that players achieve optimal exposures for match-play preparation and avoid inappropriate changes in load or accumulation of total load that can heighten injury risk or mitigate positive physical adaptation (Sombelon et al., 2017). This could be compounded by coaching practices within tennis that result in more blocked approaches to serve training approaches towards the end of sessions (Fernandez-Fernandez et al., 2021). Accordingly, monitoring of serve volume and intensity could be used to infer the mechanical stresses imposed at the upper limb and trunk (Manske et al., 2021) and better manage player preparation strategies and health outcomes. However, the availability of such insights has been constrained by the availability of wearable technology to report tennis-specific metrics in a practical and cost-effective manner.

Prescribing appropriate serve training loads requires both volume and intensity measures to understand the magnitude and distribution of exposure within practice and match-play contexts (Pluim et al., 2023). Indeed, match-play notation reveals an average of 40 ± 8 serves per set (≈ 80 per match) in national-level junior and professional females (Myers et al., 2016; Perri et al., 2018). However, this manual notation is laborious for staff and lacks insights on serve intensity outside of additional implementation of resource-intensive recording of ball speed from radar technologies or computer vision technologies (Gescheit et al., 2015). Combined volume and intensity reports for females, using Hawk-Eye technology at junior and professional Grand Slams data, describe similar serve counts of 64–80 serves per match at median speeds of $143\text{--}156\text{ km.h}^{-1}$ (Kovalchik & Reid, 2017), though this technology is cost-prohibitive in most training contexts. These limitations complicate longitudinal data collection and contributes to the sport's lack of understanding regarding the volume and intensity of serving in both training and match-play contexts.

Developments in recent tennis literature point to the increased uptake of wrist- and trunk-worn wearable devices (e.g. Sony Smart Tennis, Armbeep, Catapult) for detecting tennis strokes, especially the serve, where detection accuracy approximates 98–99% (Myers et al., 2019; Perri et al., 2022a). Such technology represents a more practical and contemporary load monitoring strategy, particularly for shots like the serve. For example, the junior male populations using wearable microtechnology in training have been shown to serve 54 ± 34 times per session (Genevois et al., 2018), with the highest relative serve count and accelerometer load recorded in technical drills (i.e. blocked serving) (Perri et al., 2023). Whilst this type of serving pattern (i.e. distribution and density of practice) is not representative of match-play (Fernandez-Fernandez et al., 2021; Filipcic et al., 2023), it is rarely quantified or reported to inform player preparation of a critical part of the game. Hence, monitoring the serve through wearable technology could provide desirable training insights (i.e. volume, intensity, distribution) for coaches that were previously unattainable.

As an example of possible insights, increased longitudinal data capture of serve accumulation with wearable microtechnology may translate into more detailed understandings of serving periodisation prior tournaments. Indeed, the $\approx 200\text{--}250$ cumulative weekly serves reported by Genevois et al. (2018) represents a useful starting point in the literature, but is limited to a single microcycle from 14 adolescent males (mean age $15.4 \pm 2\text{y}$). Alternatively, serving actions contribute $\approx 26\%$ of total shots during professional match-play (Whiteside & Reid, 2017) and indicates training exposures could be based on relative

serve volumes, though has not been examined in the current literature. Accordingly, the aims of this study were to describe and compare the training load characteristics (volume, intensity, distribution) of the serve in training and tournament mesocycles. Further, within- and between-mesocycle comparisons of serve metrics for session type (i.e. practice vs. match-play) and drill type (i.e. serve-focused drills vs. point-play) were analysed. A secondary aim of this study was to compare cumulative weekly serve counts between training weeks prior to tournaments.

2. Materials and methods

2.1. Participants

An elite squad with seven female tennis players (18.8 ± 2.9 y) volunteered for this study. The players were part of Tennis Australia's National Academy programme, who were on full-time junior ($n = 4$) or professional ($n = 3$) agreements. Training weeks typically consisted of ≈ 20 h of on-court training and ≈ 6 h of off-court training activities. Concurrently, players eligible for international junior tournaments were competing in these events regularly and alongside entry-level professional events. All junior players were ranked inside the International Tennis Federation (ITF) top 100, with professional players ranked from 185 to 1160 on the Women's Tennis Association (WTA) tour. These schedules resulted in a mixture of 17–23 annual domestic and international events per participant. As part of Tennis Australia's supported athlete agreements, players consent to their data collected and its de-identified use in potential research projects, with parental consent required for all players under the age of 18. The University's human research ethics committee approved the study (ETH23–7974).

2.2. Data collection

An observational study design was implemented, where accelerometer data from global positioning systems (GPS) was collected during training and tournament mesocycles throughout the 2022 season (March to December). Classification of training and tournament mesocycles were made in accordance with previous tennis research (Perri et al., 2022). Training mesocycles only consisted of practice sessions, whilst tournament mesocycles contained practice sessions and official matches. Within dedicated training mesocycles, all practice sessions were delivered by high-performance coaches within the National Academy, who had playing and coaching careers spanning >10 years. Sessions were conducted on a mixture of indoor and outdoor hard courts in Australia and internationally. Practice sessions during tournament mesocycles were mostly organised by the individual players in absence of coach support and occurred on match- and non-match days. Official matches were contested on hard courts under ITF junior girls and professional women's events, where wearable technology is permitted during both singles and doubles match-play. Table 1 presents the number of observations under practice and match-play conditions. Drills during practice sessions were classified into common categories as part of the daily athlete monitoring systems (Murphy et al., 2014; Perri et al., 2023), with drills determined as being part of the practice or match “hit up”

Table 1. Mean $\pm$ SD of serve count (n), serve load (AU) and serve intensity (rps) during practice sessions and official matches in training and tournament mesocycles.

Total Session Count (n)	Training Mesocycle	Tournament Mesocycle	
	294	223	
	Practice Sessions	Practice Sessions	Official Matches
Session Count (n)	294	103	120
Total Serve Events (n)	12444	4531	8332
Absolute Serve Count (n)	50 $\pm$ 27	54 $\pm$ 29	71 $\pm$ 37
[range]	[11–142]	[11–175]	[12–201]
Relative Serve Count (n.min ⁻¹)	0.77 $\pm$ 0.54	0.99 $\pm$ 0.40	0.76 $\pm$ 0.26
[range]	[0.12–4.85]	[0.10–2.21]	[0.16–1.31]
Absolute Serve Load (AU)	26 $\pm$ 16 ^b	28 $\pm$ 18 ^a	50 $\pm$ 27
[range]	[4–104]	[6–96]	[17–134]
Relative Serve Load (AU.min ⁻¹)	0.40 $\pm$ 0.28	0.49 $\pm$ 0.22	0.53 $\pm$ 0.22
[range]	[0.00–2.20]	[0.10–1.60]	[0.20–1.10]
Relative Serve Load (AU/serve)	0.51 $\pm$ 0.28	0.50 $\pm$ 0.27	0.68 $\pm$ 0.42
[range]	[0.13–4.73]	[0.27–3.44]	[0.25–4.17]
Relative Rotation Magnitude (rps/serve)	2.44 $\pm$ 0.45	2.42 $\pm$ 0.46	2.41 $\pm$ 0.40
[range]	[0.93–5.56]	[1.59–4.56]	[1.24–4.47]

AU = arbitrary units; rps = revolutions per second.

^alarge effect size ($d > 0.8$) for within-mesocycle differences between sessions.

^blarge effect size ($d > 0.8$) for differences between training sessions and tournament matches.

or “warm up” removed from the analysis. Accordingly, drills targeting the serve in either a blocked or distributed (i.e. “serve +1”) version were classified as “Serve-focused drills” for the purpose of this study. More specifically, serve-focused drills were those that involved “blocked” serving practice or non-point-based drill scenarios that included serving components, whereas point-play drills involved either scoring or tactical constraints or full match-play simulation. Additionally, point-play drills required serving to initiate the point starting to be included. Lastly, each week within a training mesocycle was categorised according to the time before the next tournament commenced (i.e. Week -3 = three weeks prior to the tournament start date).

All players wore a commercial GPS device (Catapult Vector S7, Catapult Sports, Melbourne, Australia), with in-built accelerometer, gyroscope and magnetometer sampling at 100 Hz. The device was worn in a neoprene harness, with a combined weight of 102 g, that houses the unit close to the skin at the level of the distal cervical spine. Proprietary algorithms from the manufacturer have been developed to automatically detect tennis stroke and movement actions via feature extraction of accelerometer, gyroscope and magnetometer data. The precision of these prototype algorithms to detect serve events in tennis have been validated at 98–99% accuracy during match-play from manufacturer investigations and published research (Perri et al., 2022a, 2022b). For this study, three key serving variables were obtained: (1) serve count, (2) serve load and (3) rotation magnitude. Serve counts were determined via the event detection capability of prototype algorithm and are reported alongside an accelerometer load (i.e. serve load) and gyroscope load (i.e. rotation magnitude). Serve load is determined by manufacturer software and acts as a sport-specific measure of player load (smoothed sum of raw acceleration values in the x, y and z planes), measured in arbitrary units (AU) (Bredt et al., 2020), whilst rotation

magnitude provides a measure of stroke intensity from the instantaneous gyroscope value and measured in revolutions per second (rps).

All outcome measures were reported in absolute values across practice sessions during both training and tournament mesocycles as well as official match-play during tournament mesocycles. For each session type, serve count was normalised for time (i.e. per minute [min^{-1}]) to determine relative serve count as a proxy measure of serving distribution. Additionally, serve load and serve magnitude were normalised for both time and individual serve (i.e. per serve [$/\text{serve}$]). Drill-level data followed the same pattern across both serving and point-play drills, though were not normalised on a per serve basis due to available data. Cumulative weekly serve counts were only reported during dedicated training mesocycles and included weeks where five or more days were available for analysis. This led to the final three weeks of a given training mesocycle being used in the weekly microcycle analyses.

2.3. Data processing

Raw data from the GPS unit was extracted using manufacturer software (Catapult OpenField, 3.4.99 Beta Build #74009, Catapult Sports, Melbourne, Australia) and aligned to periods of specific drills or match-play activities. Following the identification of drills or match-play periods within a given session, data from the manufacturer's prototype tennis algorithm was available via a cloud-based Shiny app. All data from the algorithm were exported as comma separated values (.csv) files, with data cleaning performed in the R language (RStudio, 4.2.2, RStudio, Inc.). The cleaned data files were then imported into Tennis Australia's cloud-based athlete management system (Smartabase, 6.13.5, Brisbane, Australia) in accordance with the consent provided in the athlete scholarship agreements. This completed dataset was assessed for outliers across the serve load and rotation magnitude metrics. Specifically, data for all serving metrics were visualised using box and whisker plots alongside the function "*boxplot.stats*" in R (RStudio, 4.2.2, RStudio, Inc.), which identified rotation magnitudes greater than 5 rps ($n = 3$) could be classified as outliers and were removed from the analysis. Additionally, to mitigate the influence of false positive serve events being recorded in training sessions or drills and contribute to the final analysis, session- and drill-level data were filtered to remove observations with < 10 serves.

2.4. Statistical analysis

All statistical analysis was performed in the R language using RStudio (RStudio, 4.2.2, RStudio, Inc.). Descriptive statistics of the mean and standard deviation were used for reporting serve count, serve load and rotation magnitude across conditions. These comparisons included between session-type differences within mesocycle-type and between-session differences across training and tournament mesocycles. Drill-level comparisons were determined within-mesocycle, with between-mesocycle comparisons for serving and point-play drills also reported. Within training microcycles, weekly serve count for each training week was analysed against each other to determine week-to-week variation. Due to the small sample size of this study, effect size (ES) using Cohen's d statistic with 95% confidence intervals (CI) were calculated to compare serve metrics across conditions. Effect sizes were interpreted according to previous research with $d < 0.2$ classified as trivial, $d = 0.2\text{--}0.5$ as small, $d = 0.5\text{--}0.8$ as medium and $d > 0.8$ as large effects.

3. Results

Table 1 presents the serve count, serve load and rotation magnitude metrics for practice sessions and official match-play during training and tournament mesocycles across all players included in the sample. Medium ES ($d = 0.51\text{--}0.66$ [95% CI = $0.25\text{--}0.87$]) existed for greater absolute serve counts during official matches versus practice sessions. Trivial ES was observed for absolute serve count between practice sessions in training and tournament mesocycles ($d = 0.14$ [95% CI = $-0.08\text{--}0.37$]). For relative serve count, trivial ES ($d = 0.03$ [95% CI = $-0.25\text{--}0.18$]) existed between practice sessions during training mesocycles and official match-play, though practice sessions in tournament mesocycles demonstrated a medium ES for higher relative serve count compared to official matches ($d = 0.70$ [95% CI = $0.43\text{--}0.97$]). Additionally, small ES for lower relative serve count in practice sessions during training mesocycles were observed ($d = 0.46$ [95% CI = $0.24\text{--}0.69$]).

A large ES existed for absolute serve loads during official matches compared to practice sessions in both training and tournament mesocycles ($d = 0.97\text{--}1.07$ [95% CI = $0.69\text{--}1.30$]; **Table 1**). Trivial ES in absolute serve load existed between training and tournament mesocycles for practice sessions ($d = 0.10$ [95% CI = $-0.12\text{--}0.32$]). Normalising serve load for time (i.e. AU.min⁻¹) revealed a medium ES for greater serve load in official matches compared to practice sessions during training mesocycles ($d = 0.54$ [95% CI = $0.32\text{--}0.75$]), though was reduced to trivial ES when comparing practice sessions within tournament mesocycles to official matches ($d = 0.19$ [95% CI = $-0.07\text{--}0.45$]). Lastly, analysis of practice sessions showed small ES for increases in relative serve load during tournament mesocycles (AU.min⁻¹) ($d = 0.38$ [95% CI = $0.15\text{--}0.60$]). When serve load was analysed per individual serve (i.e. AU/serve), small to medium ES were evident for greater relative serve load in official matches compared with practice sessions ($d = 0.45\text{--}0.50$ [95% CI = $0.42\text{--}0.53$]). Comparisons between mesocycles showed trivial ES for relative serve load in practice sessions ($d = 0.05$ [95% CI = $0.02\text{--}0.09$]). The final analysis of session-level data showed trivial differences in rotation magnitude per serve across all conditions ($d = 0.03\text{--}0.06$ [95% CI = $-0.01\text{--}0.09$]).

Drill-level data, categorised by serve-focused or point-play drills, are reported for training and tournament mesocycles in **Table 2**. Within training mesocycles, large effects existed for greater serve count during point-play drills ($d = 0.82$ [95% CI = $0.54\text{--}1.10$]; **Table 2**). Comparisons for drill type within tournament mesocycles demonstrated small effects for higher serve count in point-play drills versus serve-focused drills ($d = 0.47$ [95% CI = $0.02\text{--}0.97$]). Within drill type comparisons between training and tournament mesocycles revealed a small ES for serve count ($d = 0.27\text{--}0.29$ [95% CI = $0.04\text{--}0.72$]). When serve count were normalised for time (i.e. n.min⁻¹), large ES were observed for increased relative serve count in serve-focused drills compared to point-play in both training and tournament mesocycles ($d = 1.10\text{--}1.87$ [95% CI = $0.59\text{--}2.19$]). However, comparisons between drill type between training and tournament mesocycles showed a small-medium ES for greater relative serve count in serve-focused drills during training mesocycles and point-play drills in tournament mesocycles ($d = 0.44\text{--}0.52$ [95% CI = $0.00\text{--}0.89$]).

Table 2 also presents the serve load and rotation magnitude measures for serve-focused and point-play drills in training and tournament mesocycles. Training and tournament mesocycles showed large ES for greater absolute serve load in point-play drills compared to serve-focused drills ($d = 0.85\text{--}1.10$ [95% CI = $0.33\text{--}1.34$]). Trivial ES existed for absolute

Table 2. Mean $\pm$ SD of serve count (n) and serve load (AU) during serve-focused drills and point-play drills in training and tournament mesocycles.

Total Drill Count (n)	Training Mesocycle 235		Tournament Mesocycle 81	
	Serve-Focused Drills 154	Point-Play Drills 81	Serve-Focused Drills 22	Point-Play Drills 59
Total Drill Count (n)				
Absolute Serve Count (n)	25 $\pm$ 11 ^a	40 $\pm$ 23	28 $\pm$ 13	34 $\pm$ 14
[range]	[11–63]	[11–105]	[11–62]	[11–71]
Relative Serve Count (n.min ⁻¹)	2.77 $\pm$ 1.16 ^a	1.14 $\pm$ 0.40	2.30 $\pm$ 0.92 ^a	1.42 $\pm$ 0.64
[range]	[0.50–5.00]	[0.40–2.30]	[0.90–4.60]	[0.40–3.40]
Absolute Serve Load (AU)	12 $\pm$ 6 ^a	24 $\pm$ 16	13 $\pm$ 6 ^a	18 $\pm$ 7
[range]	[4–32]	[5–79]	[4–25]	[6–41]
Relative Serve Load (AU.min ⁻¹)	1.34 $\pm$ 0.63 ^a	0.69 $\pm$ 0.24	1.10 $\pm$ 0.55	0.76 $\pm$ 0.30
[range]	[0.30–4.50]	[0.20–1.30]	[0.40–2.40]	[0.30–1.50]
Rotation Magnitude (rps/drill)	2.48 $\pm$ 0.39	2.48 $\pm$ 0.48	2.71 $\pm$ 0.64	2.39 $\pm$ 0.42
[range]	[1.84–3.61]	[1.72–3.68]	[2.01–3.90]	[1.81–4.00]

AU = arbitrary units; rps = revolutions per second.

^alarge effect size ($d > 0.8$) for between-drill comparisons within mesocycles.

serve load during serve-focused drills between training and tournament mesocycles ($d = 0.09$ [95% CI = -0.36 – 0.53]). Small ES were observed for reduced absolute serve load in point-play drills during tournament mesocycles compared to training mesocycles ($d = -0.50$ [95% CI = -0.84 – -0.16]). Normalising serve load for time (i.e. AU.min⁻¹) showed large ES towards increased relative serve load during serve-focused drills within training mesocycles ($d = 1.35$ [95% CI = 1.06 – 1.65]). For tournament mesocycles, this similar pattern showed a medium ES ($d = 0.75$ [95% CI = 0.25 – 1.26]). Comparing relative serve load within-drill type but between mesocycles showed small ES ($d = 0.24$ – 0.42 [0.03–0.86]). The mean rotation magnitude per drill showed trivial differences between drill type in training mesocycles ($d = 0.02$ [95% CI = -0.25 – 0.29]). Tournament mesocycles showed medium ES for lower rotation magnitude during point-play drills ($d = 0.59$ [95% CI = 0.10 – 1.09]). Serve-focused drills had a small ES for increased rotation magnitude during tournament mesocycles ($d = 0.43$ [95% CI = 0.02 – 0.88]), whilst trivial ES were observed for point-play drills ($d = 0.19$ [95% CI = -0.14 – 0.53]; Table 2).

A total of 26 weeks across the sample of seven players met the criteria for weekly analysis. Average weekly serve count for the subset of three weeks prior to tournament commencement (Figure 1). Weekly serve count at Week -1 showed trivial effects between Week -2 ($d = 0.12$ [95% CI = -0.83 – 1.06]) and small effects compared to Week -3 ($d = 0.25$ [95% CI = -0.76 – 1.27]), with Week -3 showing trivial effects compared to Week -2 ($d = 0.15$ [95% CI = -0.76 – 1.06]).

4. Discussion

This study analysed the volume, distribution and intensity of serving during training and tournament mesocycles and compared between practice sessions and official matches as well as across specific serving and point play drills within respective training and tournament mesocycles. Secondly, this study also compared the weekly change in cumulative serve count prior to tournaments. Session-level data showed that serve load appears to be the key metric that increases during official match-play and infers greater

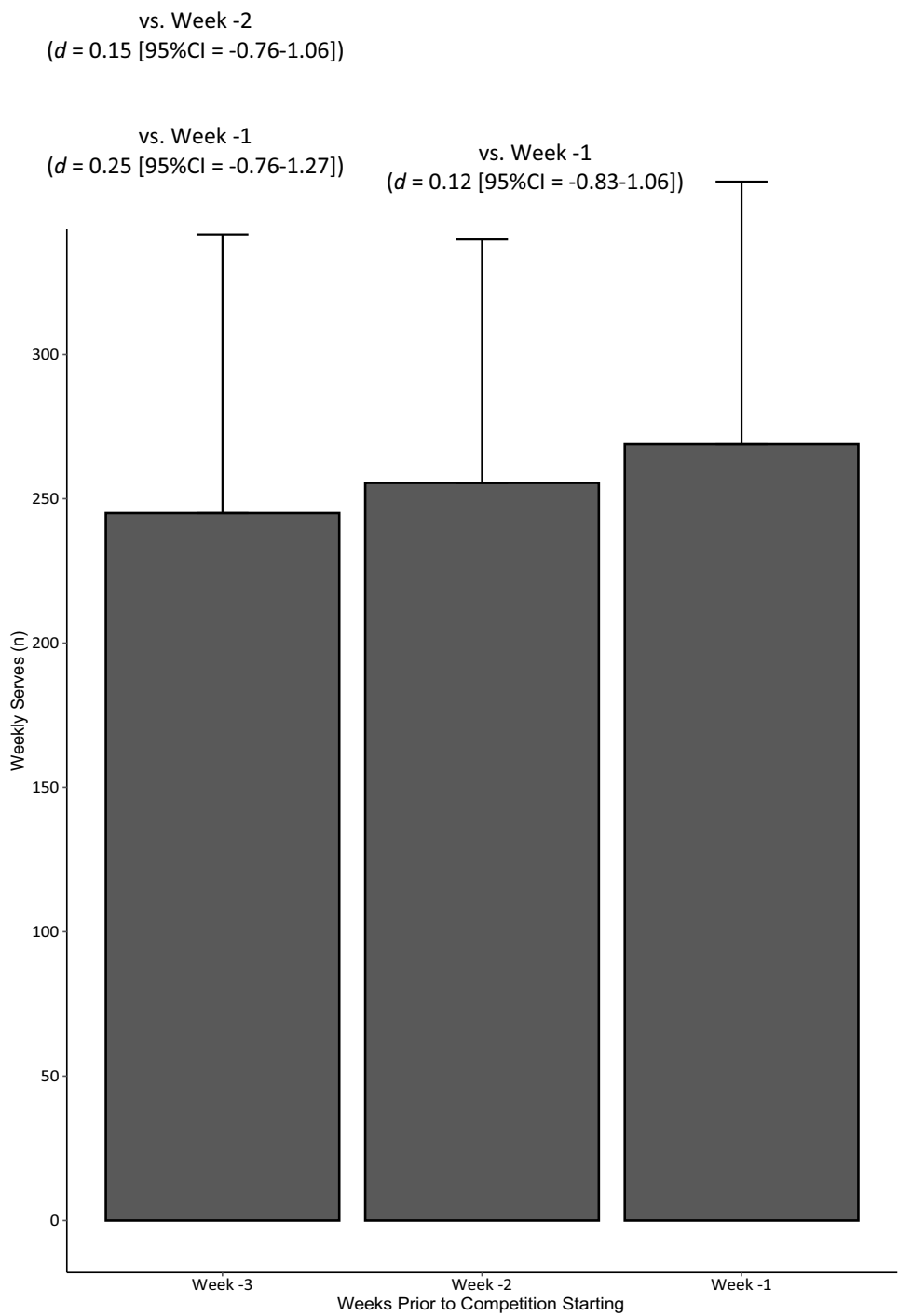


Figure 1. Mean $\pm$ SD of weekly serve count (n) in training mesocycles and reported for the three weeks prior to tournament commencement.

mechanical demands than those experienced in practice sessions. At the drill-level, serving distribution becomes increasingly dense during serve-focused drills compared to point play – a common theme reported in coach observations of tennis training patterns (Fernandez-Fernandez et al., 2021; Filipcic et al., 2023), yet previously unsupported empirically. Similarly, the trivial-small changes in weekly serve count is consistent with suggestions that limited variation exists in weekly training loads for tennis (Crespo, 2011; Perri et al., 2022). This first step towards longitudinal monitoring of tennis training loads, with a focus on the serve, could assist future research efforts and support player preparation strategies. These understandings could also extend to coaching perceptions on training load modification strategies in tennis as it relates to serving practice.

The volume of serving in practice sessions versus match-play has long been contemplated by coaches to determine whether their training prescription replicates competition demands (Filipcic et al., 2023; Myers et al., 2016). Indeed, the serve counts from matches in our study are similar to previous reports in junior and professional players (Kovalchik & Reid, 2017) and reinforce evidence on discrepancies between serve volume in match-play and practice sessions (Perri et al., 2018). This is an expected finding in the context of other sporting literature (e.g. soccer), where key training load metrics (e.g. distance covered) are considerably greater during matches compared to individual training sessions within the competitive period (Kelly et al., 2020). At the same time, absolute serve load showed large ES in match-play and highlights the sizeable mechanical demands imposed on the upper limb and trunk during match-play. This could highlight possible risks to player health when practice sessions fail to replicate the physical demands of match-play given the shoulder has been reported as the most frequently injured site (5.1 ± 1.1 injuries per year) in professional women's competition (Gescheit et al., 2017). With this in mind though, rotation magnitude per serve, which would be expected to better reflect upper limb and trunk intensity from serving, was unchanged across practice sessions and match-play conditions. Whilst speculative, this may provide an alternate explanation for the heightened accelerometer load attributed to lower-limb mechanics during serving (Colomar et al., 2022).

Training mesocycles in tennis are known for their atypical lengths (Fernandez-Fernandez et al., 2015), meaning that appropriate drill selection to achieve desired training loads increases in importance. The present results reveal how point-play drills in training mesocycles elicit greater serve volume and absolute accelerometer demands, likely a result of their longer duration. However, serve-focused drills invoke a less distributed pattern of serving actions, which was anticipated given that technical drills have previously shown the highest relative serve count (Perri et al., 2023). This would appear to be a useful metric obtained from wearable sensors, as experts have advised that over-prescription of serving drills in a short space of time may have a negative influence on training quality due to limited opportunities to practice returning or match-specific contexts (Buszard et al., 2017; Fernandez-Fernandez et al., 2021; Filipcic et al., 2023; Krause et al., 2019).

Monitoring the training load completed across weekly microcycles informs the level of variation needed and whether training targets have been met (Chena et al., 2021; Clemente et al., 2020). For the most part, our findings on weekly serve count would support previous tennis literature that demonstrated limited variations in weekly training load (Perri et al., 2022). The present results nevertheless showed small effects for

increased serve count at Week -1 compared to Week -3 and may reflect the desire of coaches to hone more match-play specific drills (which involve the serve) immediately before competitive periods (Morris, 2005; Porta & Sanz, 2005). However, the overall low variation in serve count could be suboptimal if applied to extended training mesocycles (e.g. 4–6 weeks). Specifically, continual exposure to high serving volumes could increase mechanical stress and resultant injury risk at the lumbar spine (Alway et al., 2019), whilst low training exposures likely limit performance gains (Vescovi, 2017) and discount the potential value in training principles like functional overreaching. Speculatively, appropriate serving exposure may be achieved through ample variation in daily serve volume, which could counteract potential injury risk (Pluim & Drew, 2016). Alternatively, benchmarking serve exposures through proportion of serves to total strokes during practice sessions may also assist match-play preparation strategies.

4.1. Limitations

Our study supports continued longitudinal reports of tennis training loads using wearable devices to provide practitioners with objective training insights, leading to more informed planning and athlete preparation. However, it is important to acknowledge certain limitations when interpreting the present findings. Indeed, the small sample ($n = 7$) of female players observed in this study represents a limitation as the findings may not accurately represent the wider women's tennis cohort or differences between junior and professional players. Other limitations of the study relate to the data analysis protocol, such as the arbitrary threshold of at least 10 serves being required in an individual session or drill. Additionally, due to missing data given the longitudinal nature of the study, the weekly periodisation analysis was limited to a three-week period prior to tournaments commencing and may not be a true reflection of serve planning within training mesocycles. Such a limitation was compounded by the stochastic training and tournament schedules in tennis, where information from travelling players were not always known to the research team. This could also include weeks where the coaching staff were controlling player on-court activities to a greater degree due to injury or illness concerns. It is also acknowledged that the court surface, which was not always known to the research team, may have influenced the training and tournament profiles of serving practice and may be a limitation of our findings. Lastly, despite the high accuracy of the algorithm (>99%) to detect serve events, it is noted that the training of this algorithm was from male players and may have altered precision for females.

5. Conclusions

This study contributes to the empirical knowledge of the tennis serve in female players through implementation of wearable microtechnology and appears to support tennis anecdotes in that the serve is typically practiced at volumes lower than match-play expectations. Further, official matches showed the highest accelerometer load, which may be useful in establishing benchmarks for training design. At the drill-level, coaching anecdotes can be supported through reporting of relative serve count as a means to understand serving distribution during practice. Whilst cumulative weekly serve counts

during training were subject to limited variation, this could invoke a more proactive approach to serve preparation strategies from coaches.

Disclosure statement

The authors would like to declare that Thomas Perri, Machar Reid and Alistair Murphy are employed by Tennis Australia.

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